17 March 2026

Exercise 4: Joins, UNION, Filtering and Aggregate

1.
```
SELECT A.EmployeeID,
       A.FirstName,
       A.LastName,
       A.Department,
       A.Salary,
       B.ProjectID,
       B.Budget,
       B.Status
FROM Employees AS A
INNER JOIN Projects AS B
ON A.EmployeeID = B.EmployeeID;
```

EmployeeID	FirstName	LastName	Department	Salary	ProjectID	Budget	Status
1	John	Doe	IT	70000	103	75000	Pending
2	Alice	Smith	HR	60000	102	50000	Ongoing
3	Bob	Johnson	Finance	75000	104	90000	Ongoing
5	Emma	Wilson	Sales	65000	105	65000	Completed
6	Michael	Clark	Finance	80000	106	80000	Pending
1	John	Doe	IT	70000	101	100000	Completed

2.

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2. SELECT A.EmployeeID,
A.FirstName,
A.LastName,
A.Department,
A.Salary,
B.ProjectID
B.ProjectName
B.Budget
B.Status
FROM Employees AS A
LEFT JOIN Projects AS B
ON A.EmployeeID = B.EmployeeID;

EmployeeID	FirstName	LastName	Department	Salary	ProjectID	ProjectName	Budget	Status
1	John	Doe	IT	70000	103	Cybersecurity	75000	Pending
2	Alice	Smith	HR	60000	102	Emp Training	50000	Ongoing
3	Bob	Johnson	Finance	75000	104	Financial Analy	90000	Ongoing
4	David	Brown	IT	72000	NULL	NULL	NULL	NULL
5	Emma	Wilson	Sales	65000	105	Market Expan	65000	Completed
6	Michael	Clark	Finance	80000	106	Risk Mngt	80000	Pending
1	John	Doe	IT	70000	101	AI Developm	100000	Completed

3. SELECT B.ProjectID,
B.ProjectName,
B.Budget,
B.Status,
A.EmployeeID,
A.FirstName,
A.LastName,
A.Department,
A.Salary

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```
FROM Employees AS A
RIGT JOIN Projects AS B
ON A.EmployeeID = B.EmployeeID;
```

Project ID	Project Name	Budget	Status	Employee ID	First Name	Last Name	Department	Salary
101	AI Dev	100000	Completed	1	John	Doe	IT	70000
102	Emp training	50000	Ongoing	2	Alice	Smith	HR	60000
103	Cybersecurity	75000	Pending	1	John	Doe	IT	70000
104	Fin Analysis	90000	Ongoing	3	Bob	Johnson	Finance	75000
105	Market Exp	65000	Completed	5	Emma	Wilson	Sales	65000
106	Risk Mngt	80000	Pending	6	Michael	Clark	Finance	80000

4.
```
SELECT A.EmployeeID,
       A.FirstName,
       A.LastName,
       A.Department,
       A.Salary,
       B.ProjectID,
       B.ProjectName,
       B.Budget,
       B.Status
FROM Employees AS A
FULL OUTER JOIN Projects AS B
ON A.EmployeeID = B.EmployeeID;
```

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EmployeeID	First Name	Last Name	Department	Salary	Project ID	Project Name	Budget	Status
3	Bob	Johnson	Finance	75000	104	F. Analysis	90000	Ongoing
4	David	Brown	IT	72000	NULL	NULL	NULL	NULL
6	Michael	Clark	Finance	80000	106	Risk Mngt	80000	Pending
2	Alice	Smith	HR	60000	102	E. Training	50000	Ongoing
5	Emma	Wilson	Sales	65000	105	M. Expansion	65000	Completed
1	John	Doe	IT	70000	103	Cybersecurity	75000	Pending
1	John	Doe	IT	70000	101	AI. Dev	100000	Completed

5.
```
SELECT DISTINCT A.City AS Location
FROM Employees AS A
UNION
SELECT DISTINCT B.Status AS Location
FROM Projects AS B;
```

Location
London
New York
Los Angeles
Sydney
Toronto
Ongoing
Pending
Completed

6.
```
SELECT A.City AS Location
FROM Employees AS A
UNION ALL
SELECT DISTINCT B.Status AS Location
FROM Projects AS B;
```

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Location
New York
Los Angeles
Toronto
London
Sydney
New York
Ongoing
Pending
Completed

7.
```
SELECT EmployeeID,
       FirstName,
       LastName,
       Department,
       Salary
FROM Employees
WHERE Salary > 70000;
```

Employee ID	First Name	Last Name	Department	Salary
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72000
6	Michael	Clark	Finance	80000

8.
```
SELECT EmployeeID,
       FirstName,
       LastName,
       Department,
       Salary
FROM Employees
WHERE Department = 'IT' OR Department = 'Finance';
```

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EmployeeID	FirstName	LastName	Department	Salary
1	John	Doe	IT	70000
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72000
6	Michael	Clark	Finance	80000

9. SELECT ProjectID,
ProjectName,
Budget,
Status
FROM Projects
WHERE Status = 'Ongoing' OR Status 'Pending';

ProjectID	ProjectName	Budget	Status	
102	Employee training	50000	Ongoing	
103	Cybersecurity Audit	75000	Pending	
104	Financial Analysis	90000	Ongoing	
106	Risk Management	80000	Pending	

10. SELECT ProjectID,
ProjectName,
Budget,
Status
FROM Projects
WHERE Budget > 70000 AND Status = 'Ongoing' OR Status = 'Pending'

ProjectID	Project Name	Budget	Status
103	Cybersecurity	75000	Pending
104	Financial Analysis	90000	Ongoing
106	Risk Management	80000	Pending

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11. SELECT EmployeeID,
FirstName,
LastName,
Department,
Salary,
City
FROM Employees
WHERE City = 'New York' OR City = 'Toronto'
ORDER BY Salary DESC;

EmployeeID	FirstName	LastName	Department	Salary	City
6	Michael	Clark	Finance	80000	New York
3	Bob	Johnson	Finance	75000	Toronto
1	John	Doe	IT	70000	New York

12. SELECT EmployeeID,
FirstName,
LastName,
Department,
Salary
FROM Employees
ORDER BY Salary DESC
LIMIT 3;

EmployeeID	FirstName	LastName	Department	Salary
6	Michael	Clark	Finance	80000
3	Bob	Johnson	Finance	75000
4	David	Brown	IT	72000

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13. SELECT Sum(Salary) AS TotalSalary,
Department
FROM Employees
GROUP BY Department
ORDER BY TotalSalary DESC;

Totalsalary	Department
155000	Finance
142000	IT
65000	Sales
60000	HR

14. SELECT City,
AVG (Salary) AS AverageSalary
FROM Employees
WHERE Salary > 65000
GROUP BY City;

City	AverageSalary
New York	75000
Toronto	75000
London	72000

15. SELECT Department,
COUNT (EmployeeID) AS EmployeeCount
FROM Employees
WHERE EmployeeID > 1
GROUP BY Department;

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Department	Employee Count
HR	1
Finance	2
IT	1
Sales	1

16.
```
SELECT Status,
       COUNT(ProjectID) AS ProjectCount
FROM Projects
GROUP BY Status
HAVING ProjectCount <= 2;
```

Status	ProjectCount
Completed	2
Ongoing	2
Pending	2

17.
```
SELECT A.EmployeeID,
       A.FirstName,
       A.LastName,
       SUM(B.Budget) AS TotalProjectBudget
FROM Employees AS A
INNER JOIN Projects AS B
ON A.EmployeeID = B.EmployeeID
GROUP BY A.EmployeeID,
         A.FirstName,
         A.LastName
HAVING SUM(B.Budget) > 150000;
```

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EmployeeID	FirstName	LastName	TotalProjectBudget
1	John	Doe	175000

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